Find MVUE

Two Methods to Find MVUE

- Get a complete and sufficient statistic Y_1. Then find an unbiased estimator $\varphi(Y_1)$ which is also a function of Y_1.
- Get a complete and sufficient statistic Y_1. Get a trivial unbiased estimator $\hat{\theta}_0$. Then define $\varphi(Y_1) = \mathbb{E}(\hat{\theta}_0 | Y_1)$, which is the MVUE of θ.

Example 1

- $X_1, \ldots, X_n \overset{iid}{\sim} f(x;\theta) = \frac{2x}{\theta^2}$, $0 < x < \theta$. What is the MVUE of θ?

Example 1

Example 1

Example 2

- $X_1, \ldots, X_n \overset{iid}{\sim} Unif([-\theta, \theta])$. What is the MVUE of θ?

Example 2

Example 2

Example 2

Example 3

- Let $X_1, \ldots, X_n \stackrel{iid}{\sim} N(\theta, 1)$. What is the MVUE of $\Phi(1-\theta)$? Note that $\mathbb{P}(X \leq 1) = \Phi(1-\theta)$.

Example 3

Example 3